

Pool-Playing Robot

v2 - End-to-End Learning | April 2026

1. Vision

A robot that plays pool on a real table. Computer vision digitizes the table, an attention-based neural network selects shots (learned entirely through self-play, no hand-coded strategy), and a robotic platform executes with a cue stick.

2. Pipeline

Stage 1: Computer Vision

Camera -> ball positions in table coordinates.

- Overhead or binocular cameras on robot head
- Homography rectification from 4 table corners
- Ball detection (Hough/CNN) + color classification
- Output: same format as simulation (zero-change AI transfer)
- Required accuracy: ~0.25 inch

Stage 2: Attention-Based AI

Every ball attends to every other ball (like tokens in an LLM).

The network learns:

- "Ball #7 near rack + #4 between cue and #7 = follow through #4 to position for #7 to break rack"
- "No clean pocket path -- clip #3 thinly, hide cue behind rack"
- "Ball #9 is the key ball -- keep it near the rack for later"

Architecture: 2-4 attention layers, 500K-2M params

Input: 16 ball tokens x 12 features

Output: aim angle, speed, spin (all continuous)

Training: PPO self-play on H100, 2-6 hours

Stage 3: Robot Platform

Recommended: rolling base + dual arm

- Mobile base rolls around table perimeter
- Two arms: bridge (fixed point) + drive (linear stroke)
- Camera on head for table observation
- Prototype in Isaac Sim, then physical build
- Manipulation policy also trained with RL on H100

3. The Attention Analogy

LLM: every token attends to every other token

-> "the word pool 3 sentences ago makes cue likely here"

Pool AI: every ball attends to every other ball

-> "ball #7 near rack + ball #4 in path = strategic opportunity"

Attention weights encode:

- Obstruction patterns (which balls block which pockets)
- Cluster structure (break opportunities)
- Key ball identification (for rack continuation)
- Position play geometry (where cue ball should end up)

All discovered through self-play. No hand-coded rules.

4. Roadmap

Phase 1 (DONE): 15K MLP, CPU, C physics, browser demo

Phase 2 (NEXT): Attention network, H100 training, end-to-end

Phase 3: Computer vision prototype

Phase 4: Isaac Sim robot design + manipulation training

Phase 5: Physical robot, sim-to-real transfer

Phase 6: Calibration, real-game performance

5. Shared Infrastructure (Unitree G1)

- H100 cloud GPU
- PPO/GAE training framework (RSL-RL)
- Isaac Sim for robot prototyping
- Sim-to-real transfer methodology

G1 learns locomotion (continuous control, contacts)

Pool AI learns strategy (geometry, planning)

Both use: parallel envs, GPU acceleration, self-play

6. Technical Challenges

Sim-to-real: domain randomization (vary friction, restitution)

Aim precision: 0.1-degree error = 0.09 inch miss at 50 inches

Spin control: contact point on cue ball requires ~1mm accuracy

Table access: reaching across 4.5x9 foot surface

Scaling: same architecture works for 8-ball, 9-ball, snooker